Cairo University – Faculty of Engineering
Computer Engineering Department • CMPN451
Data Mining, Big Data and Analytics • Spring 2021

Lab 6
Linear Regression

Submitted by

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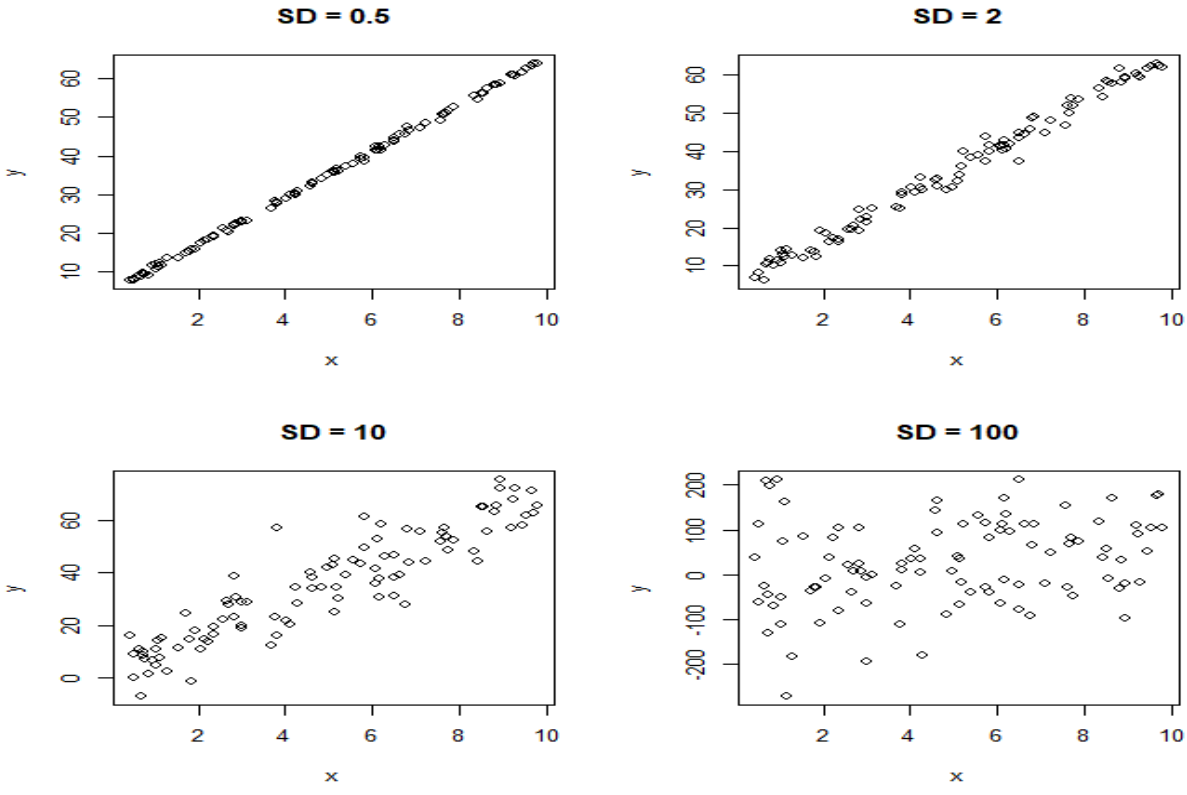
Part 1 – Synthetic Linear Data

Q1. How do data points change for different values of standard deviation?

The standard deviation (sd) controls the amount of noise added to the linear model $y = 5 + 6x + \varepsilon$, where $\varepsilon \sim N(0, sd^2)$.

- Small sd (e.g., 0.5): Data points cluster very tightly around the regression line — the linear relationship is clearly visible.
- Default sd (2): Moderate scatter around the line — the trend is still clear but with visible spread.
- Large sd (e.g., 10): Points are widely scattered and the linear relationship becomes hard to see visually.
- Very large sd (e.g., 100): Points are completely random-looking with no visible trend at all — the noise completely overwhelms the signal.

Which means that the larger the standard deviation, the further the data deviates from the linear model.



Q2. How are the coefficients affected by changing the standard deviation?

The true model has intercept = 5 and slope = 6. Changing sd affects how close the estimated coefficients are to these true values:

- Small sd → Estimated coefficients stay very close to the true values (5 and 6) because there is little noise.
- Large sd → Coefficients deviate more from the true values because the noise dominates the signal.
- Very large sd (e.g., 100) → Coefficients become very unreliable and can differ dramatically from the true values (5 and 6), since the noise completely dominates.

In all cases, the coefficients will be approximately correct on average — but their variance increases with sd.

```
sd(0.5) model coefficients:

Call:
lm(formula = y_sd0.5 ~ x)

Coefficients:
(Intercept)            x
      4.909        6.013

sd(2) model coefficients:

Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)            x
      5.012        6.015

sd(10) model coefficients:

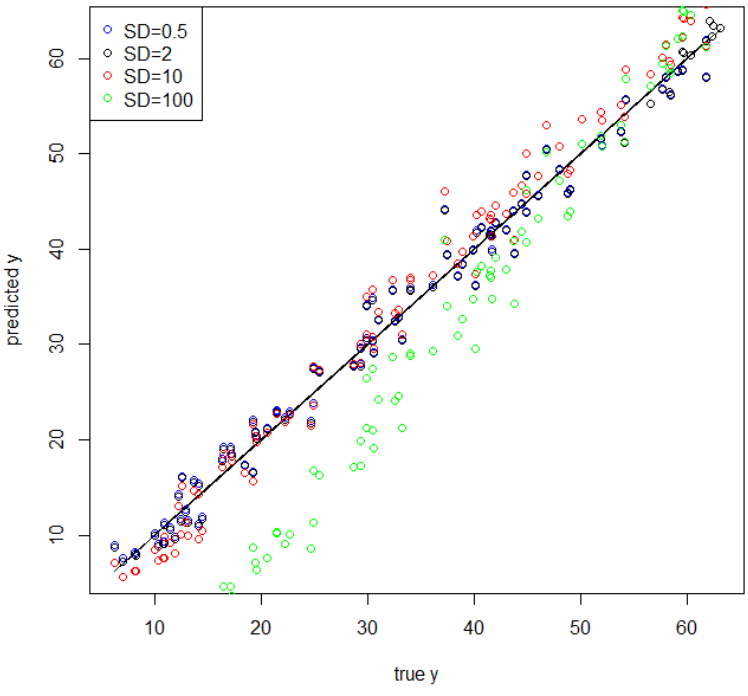
Call:
lm(formula = y_sd10 ~ x)

Coefficients:
(Intercept)            x
      7.130        5.606

sd(100) model coefficients:

Call:
lm(formula = y_sd100 ~ x)

Coefficients:
(Intercept)            x
     24.385        3.652
```



Q3. How is R-squared affected by changing the standard deviation?

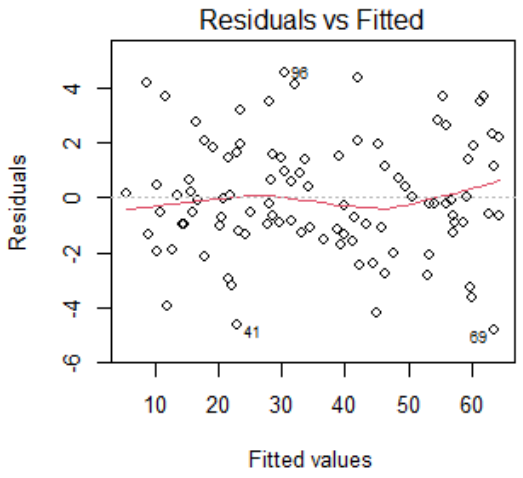
- Small sd → R-squared is close to 1.0, meaning the model explains almost all the variance in y.
- Large sd → R-squared drops down in the 0 direction, because the noise makes it hard for x to explain the variance in y.
- Very large sd (e.g., 100) → R-squared approaches 0 or may even be essentially 0, as the noise completely overwhelms any relationship between x and y.

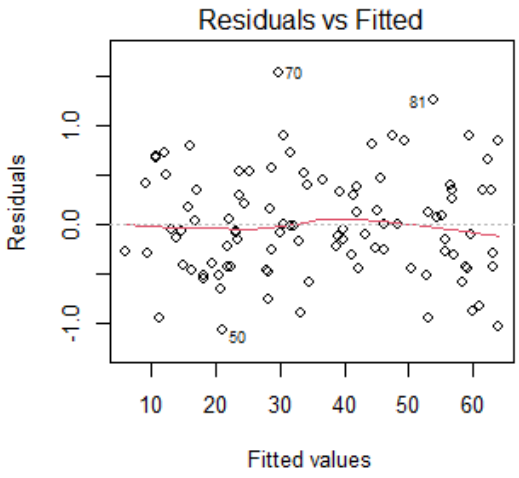
R-squared measures the proportion of variance in y explained by x. More noise = less explained variance = lower R-squared.

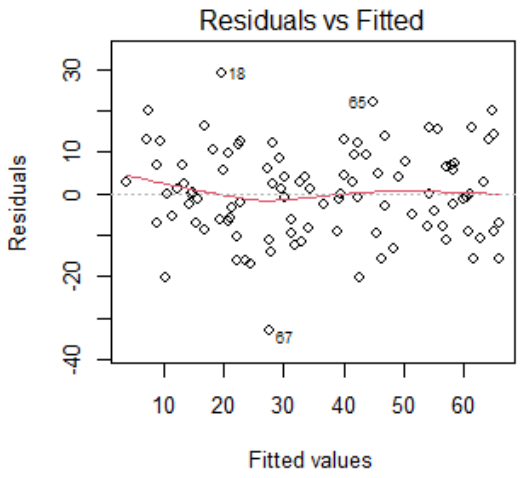
```
OLS sd(0.5) gave slope of  6.013052 and an R-sqr of  0.9992019
OLS sd(2) gave slope of  6.014534 and an R-sqr of  0.9895824
OLS sd(10) gave slope of  5.606005 and an R-sqr of  0.7330281
OLS sd(100) gave slope of  3.652102 and an R-sqr of  0.009506953
```

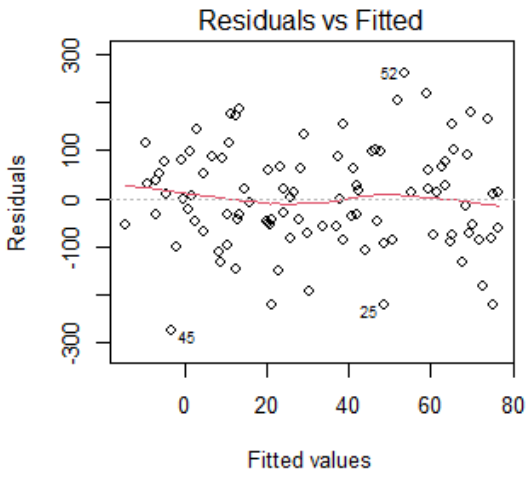
Q4. What do you conclude about the residual plot? Is it a good residual plot?

Yes, this is a good residual plot. The residuals are randomly scattered around the horizontal zero line with no visible pattern or curve, and the variance appears roughly constant across all fitted values (homoscedasticity). This confirms that the linear model is appropriate for this data.





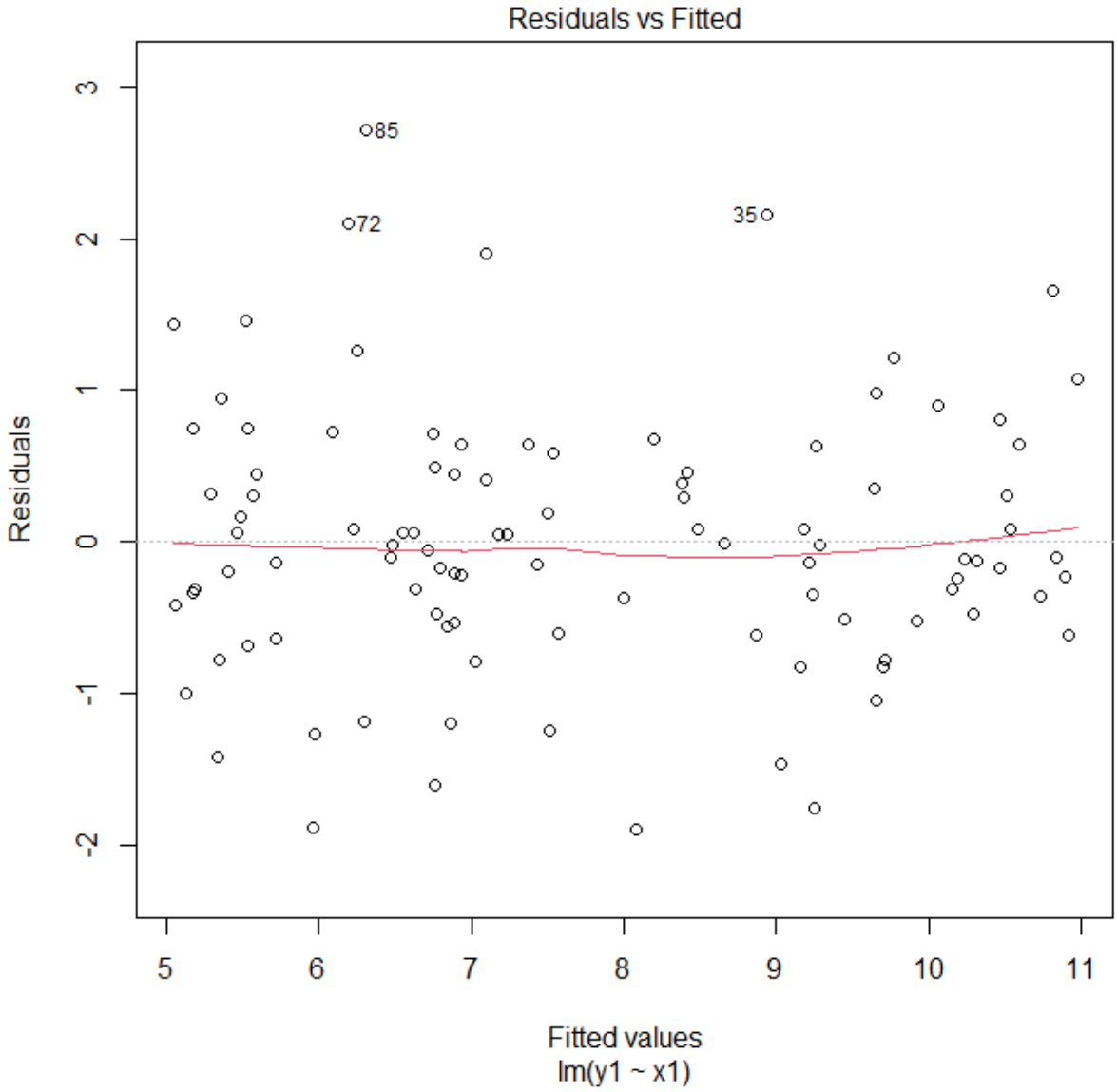




Part 2 – Nonlinear Data with Linear Model

Q5. What do you conclude about the residual plot? Is it a good residual plot?

Yes, this is still a good residual plot. The nonlinear coefficient (0.1) in the model $y = 5 + 6x + 0.1x^2 + \varepsilon$ is very small, so the nonlinearity is negligible. The residuals appear randomly scattered with no discernible pattern, and the linear model remains a good approximation in this case.

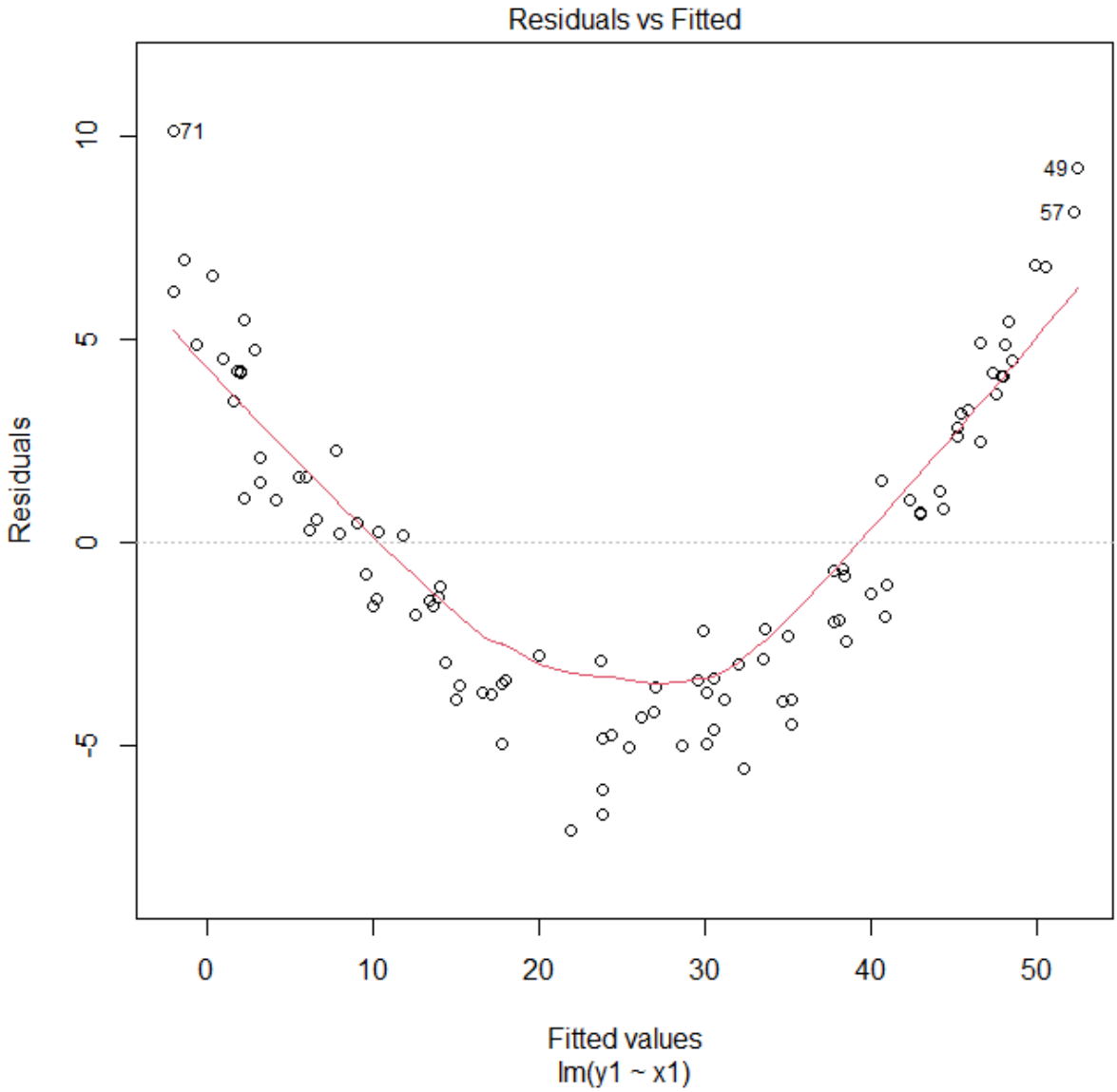


Q6. Change the nonlinear coefficient to a large value. What do you notice?

When the nonlinear coefficient is set to a large value (e.g., 50), so that the model becomes $y = 5 + 6x + 50x^2 + \varepsilon$:

- The residual plot shows a clear curved (U-shaped or inverted U-shaped) pattern.
- Residuals are NOT randomly scattered — they are systematically positive in some regions and negative in others.
- This is a bad residual plot. It indicates that the linear model is no longer adequate for this data.

Conclusion: When nonlinearity is strong, a polynomial or nonlinear model should be used instead of a simple linear regression.

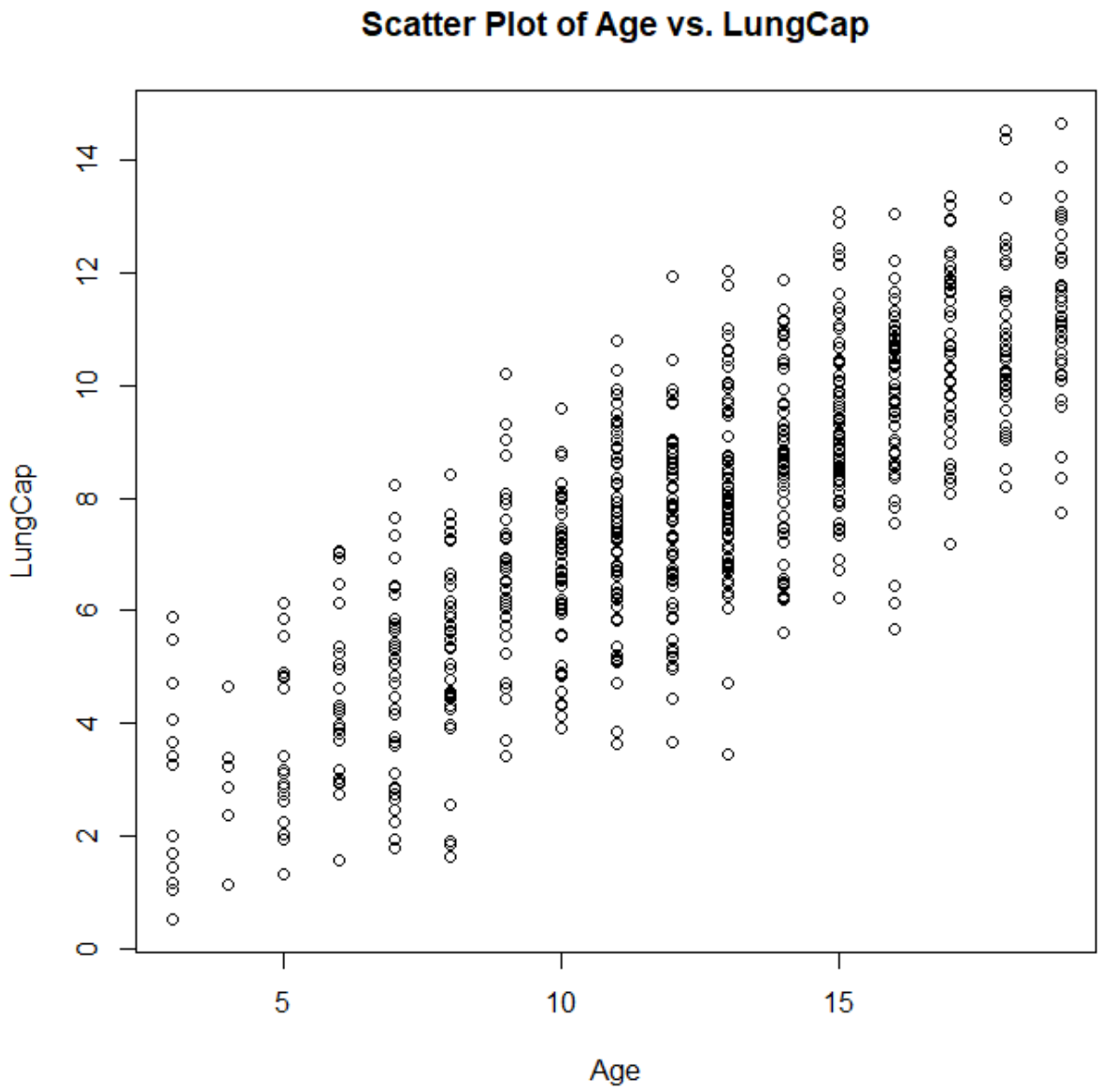


Part 3 – Lung Capacity Dataset

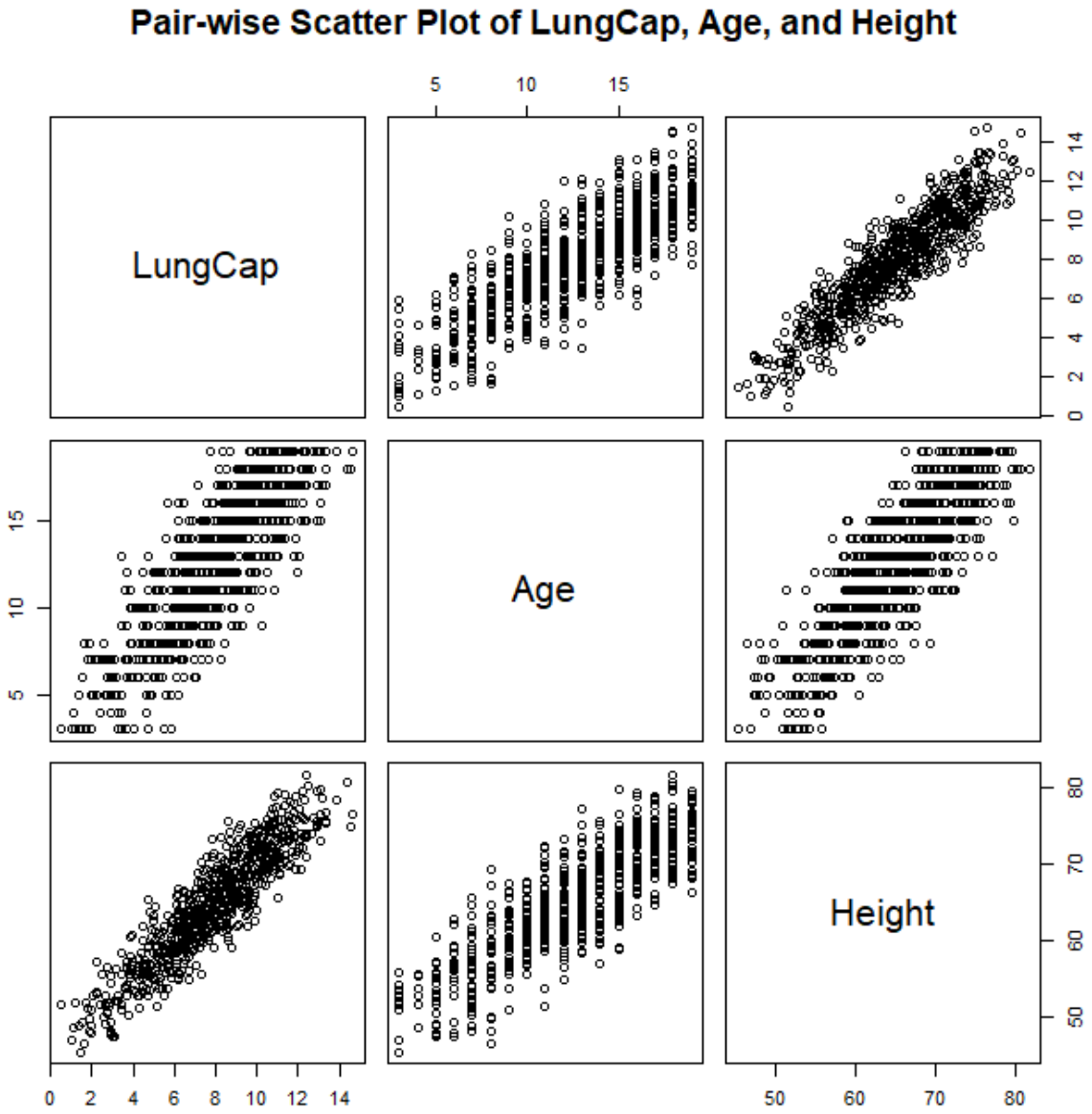
Q7. Import the dataset. What are the variables?

```
"LungCap"    "Age"        "Height"     "Smoke"      "Gender"     "Caesarean"
```

Q8. Scatter plot of Age (x-axis) vs LungCap (y-axis)



Q9. Pairwise scatter plot between LungCap, Age, and Height



Q10. Correlation between Age & LungCap, and Height & LungCap

```
Q10: Correlation between Age and LungCap:  0.8196749
Correlation between Height and LungCap:  0.9121873
```

Q11. Which variable is more correlated to LungCap?

Height is more correlated to LungCap (r = 0.9122) than Age (r = 0.8197).

Q12. Are Height and LungCap correlated? Why?

Yes, Height and LungCap are strongly positively correlated (near 1) (r = 0.9122).

This makes biological sense: taller people have larger chest cavities, which physically allows for greater lung capacity. As height increases, the lungs have more space to expand, resulting in higher lung capacity.

Q13. Fit a linear regression model with LungCap as dependent and all other variables as independent

```
model <- lm(LungCap ~ ., data=data)
# The dot (.) means: use ALL other columns as independent variables
# i.e., Age + Height + Smoke + Gender + Caesarean
```

Q14. Show a summary of the model

```
Q14: Summary of the linear regression model:

Call:
lm(formula = LungCap ~ ., data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-3.3388 -0.7200  0.0444  0.7093  3.0172

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -11.32249    0.47097 -24.041  < 2e-16 ***
Age            0.16053    0.01801   8.915  < 2e-16 ***
Height         0.26411    0.01006  26.248  < 2e-16 ***
Smokeyes      -0.60956    0.12598  -4.839 1.60e-06 ***
Gendermale     0.38701    0.07966   4.858 1.45e-06 ***
Caesareanyes  -0.21422    0.09074  -2.361   0.0185 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.02 on 719 degrees of freedom
Multiple R-squared:  0.8542,    Adjusted R-squared:  0.8532
F-statistic: 842.8 on 5 and 719 DF,  p-value: < 2.2e-16
```

Q15. What is the R-squared value? What does it indicate?

```
Q15: R-squared value:  0.8542478
```

R-squared indicates the proportion of variance in the dependent variable (LungCap) explained by the independent variables in the model. A value of 0.8542 means that **85.42% of the variance in LungCap is explained by the independent variables** (Age, Height, Smoke, Gender, and Caesarean), which indicates a strong model fit. The remaining 14.58% is unexplained variance due to other factors not included in the model.

Q16. Show the coefficients. Do they make sense?

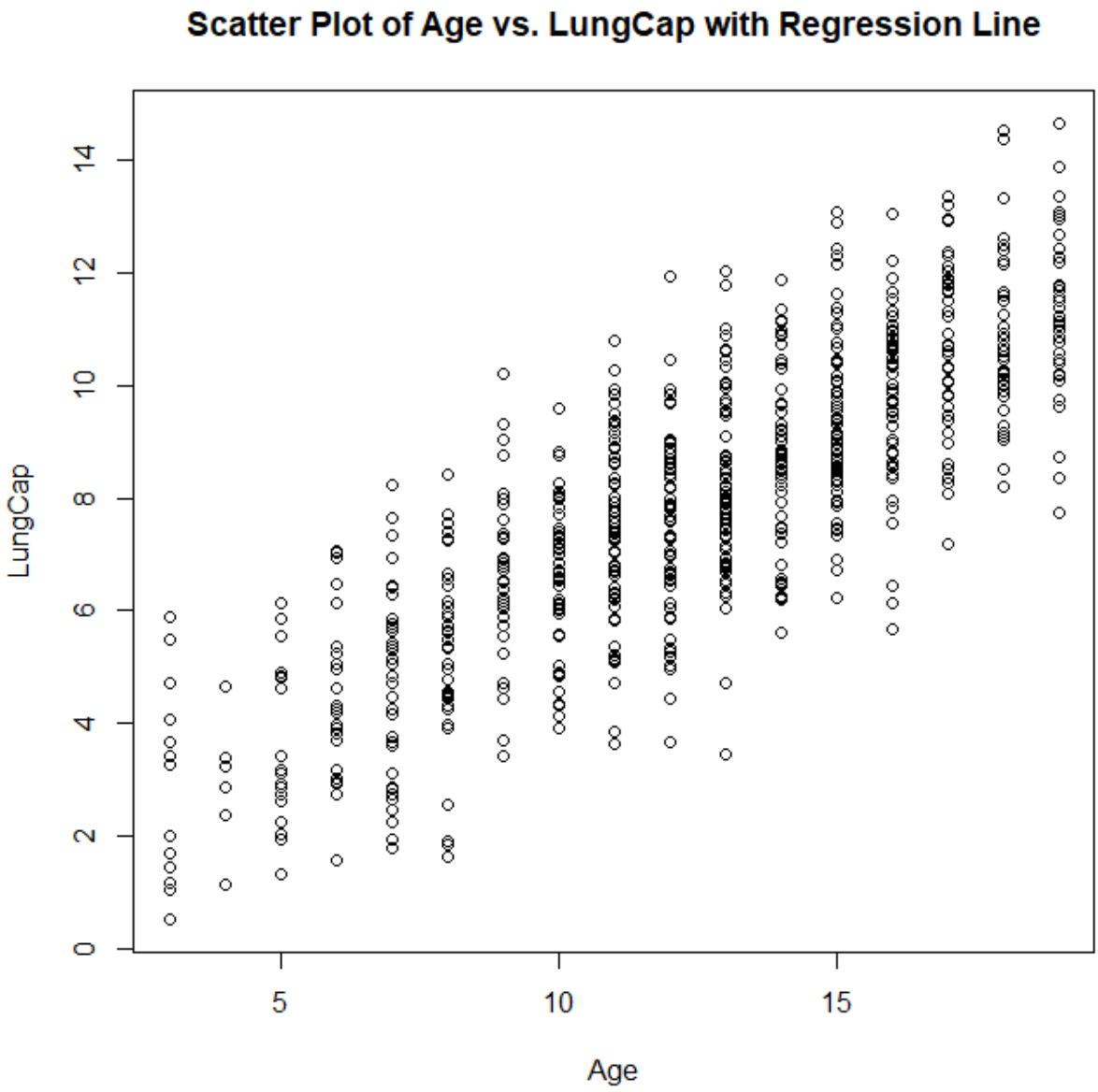
```
Q16: Coefficients of the linear model:
 (Intercept)         Age      Height    Smokeyes  Gendermale Caesareanyes
 -11.3224856   0.1605296   0.2641128  -0.6095592   0.3870117  -0.2142182
```

Interpretation of each coefficient:

- Intercept (-11.3225): Does not make sense. It implies that when all variables are zero, LungCap would be -11.32, which is physically impossible. This is because extrapolating Height and Age to zero is unrealistic. The intercept is simply a mathematical fitting constant here.
- Age (+0.1605): Makes sense — older individuals tend to have higher lung capacity.
- Height (+0.2641): Makes sense — taller people have larger lungs.
- Smoke (-0.6096): Makes sense — smoking reduces lung capacity.
- Gender (+0.3870): Makes sense — males generally have higher lung capacity than females.
- Caesarean (-0.2142): Questionable. It is not immediately obvious why being born via C-section would reduce lung capacity. This variable may not be meaningful and should be investigated further.

What to do about the questionable variable: Age and Height are highly correlated (we saw this in Q10, r = 0.82 and 0.91 with LungCap), meaning multicollinearity exists. We should consider removing one of them.

Q17. Redraw scatter plot of Age vs LungCap and overlay the regression line



Why is the line NOT displayed on the plot?

abline() only uses the first two coefficients of the model — the intercept and the coefficient of Age — and draws the line as:

LungCap = -11.3225 + 0.1605 × Age

The large negative intercept (-11.3225) pushes the entire line far below the visible range of the plot, since LungCap values in the dataset start around 0.5. So, the line exists mathematically but is drawn outside the visible plot area. And this is caused by the correlated and dependent variables.

Q18. New model using only Age, Smoke, and Caesarean

Summary for the new model:

```
Call:
lm(formula = LungCap ~ Age + Smoke + Caesarean, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-4.7428 -1.0580 -0.0490  0.9985  4.2319

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.10867    0.18419   6.019 2.79e-09 ***
Age           0.55617    0.01439  38.639  < 2e-16 ***
Smokeyes     -0.64310    0.18681  -3.443 0.000609 ***
Caesareanyes -0.14603    0.13468  -1.084 0.278610
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.514 on 721 degrees of freedom
Multiple R-squared:  0.6778,    Adjusted R-squared:  0.6764
F-statistic: 505.5 on 3 and 721 DF,  p-value: < 2.2e-16
```

Q19. Coefficients and scatter plot for Model 2

```
Q19: Coefficients of the new linear model:
 (Intercept)          Age     Smokeyes Caesareanyes
   1.1086723    0.5561667   -0.6431029   -0.1460278
```

Do they make sense?

- Intercept (+1.1087): More reasonable than Model 1's intercept, but still not fully meaningful since Age = 0 is unrealistic.
- Age (+0.5562): Makes sense — lung capacity increases with age.
- Smoke (-0.6431): Makes sense — smoking reduces lung capacity.
- Caesarean (-0.1460): Still questionable for the same reasons as before.

What happened compared to Model 1?

The coefficient of Age jumped from 0.1605 to 0.5562 when Height was removed. This is because Age and Height are correlated — when Height is absent from the model, Age absorbs its explanatory power.

For the scatter plot with the regression line from Model 2:

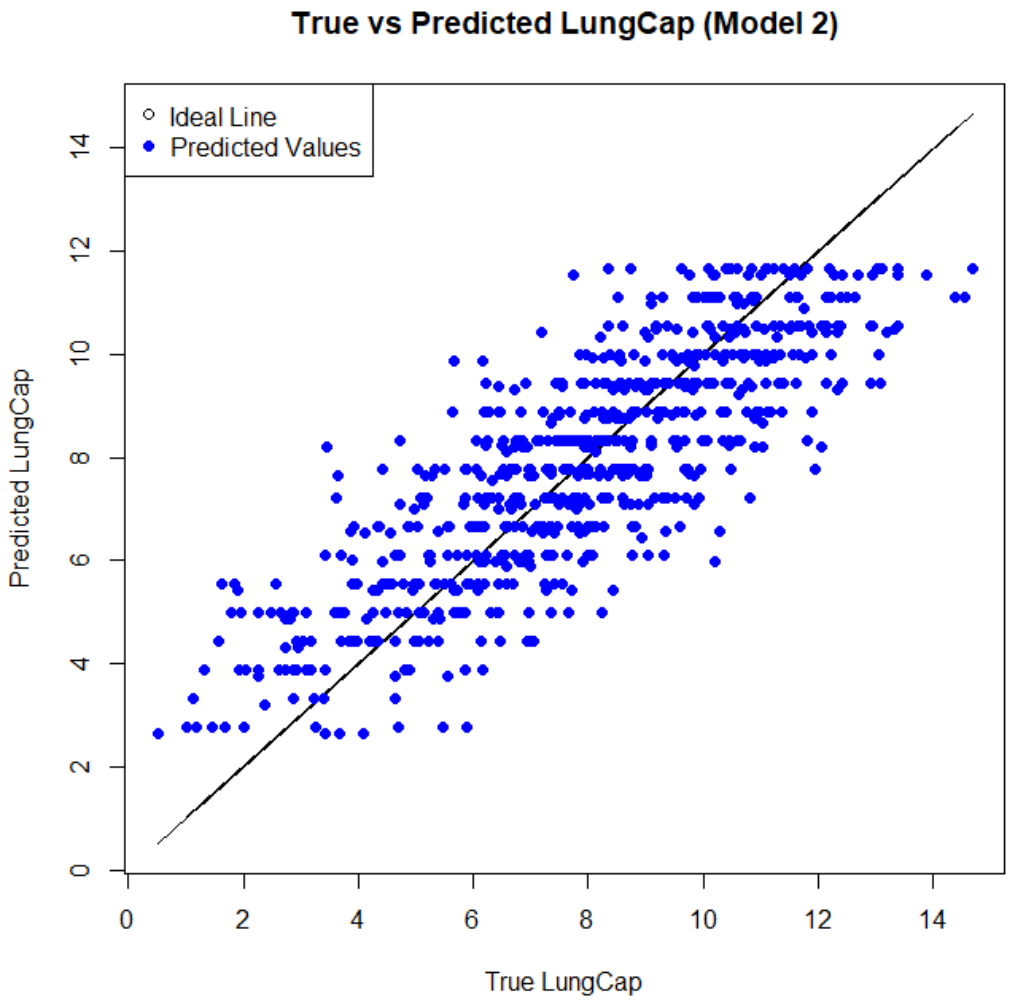


Now the line IS visible,

LungCap = 1.1087 + 0.5562 × Age → intercept is now reasonable, so the line falls within the visible range and appears correctly on the plot.

The intercept changed from -11.3225 to +1.1087 because Height was removed from the model. In Model 1, Height was pulling the intercept down significantly since it needed to compensate for the large Height values in the data. Without Height, the intercept returns to a more reasonable value.

Q20. Predict results on training data



Q21. Calculate the Mean Squared Error (MSE) of the training data

```
mse <- mean((data$LungCap - ypred2)^2)
```

```
Q21: Mean Squared Error (MSE) of the training data:  2.280169
```